UCSD ROCKET PROPULSION LABORATORY • 2024–2025

# Daedalus | Structures Lead

## Engineering objective

Design and manufacture a student rocket; 4,086 ft was the historical simulated apogee.

## My structures scope

Nose geometry, shoulder/body fit, recovery attachment and structural integration.

## Manufacturing and integration

Supported PLA structural manufacture, mechanical fit checks and hardware integration with the team.

**Native assembly • reference envelope**



**25 in**
tip to aft body

**1.5 in**
body diameter

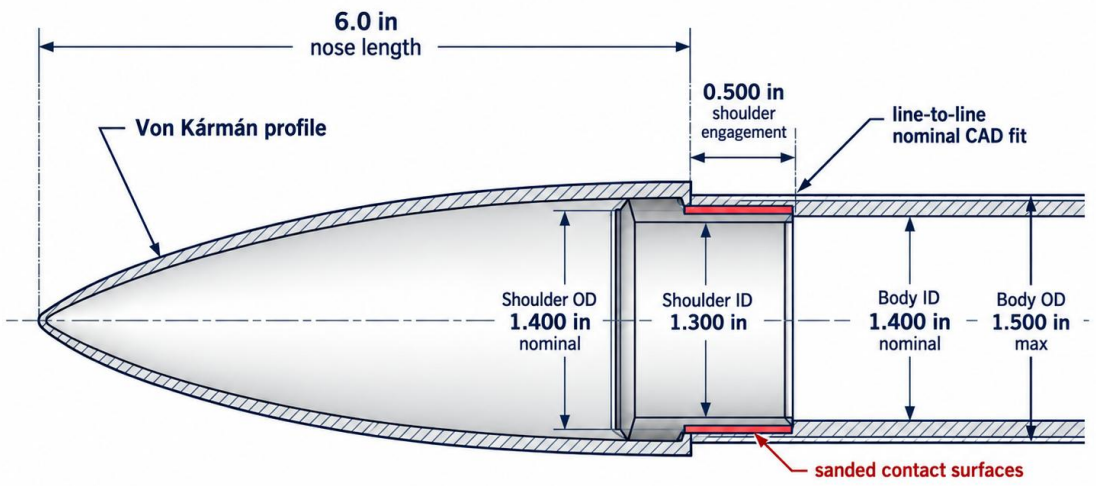
**6 in nose**

Status  Manufactured and ground-tested; never flown.  Performance figures are simulations.

Assembly image and dimensions are an October 2026 reference reconstruction, not as-built measurements.

# What carries the deployment load?

## NOSE / BODY FIT

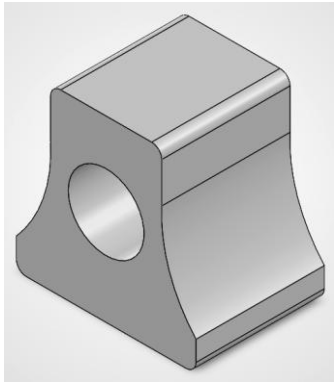



$$t_{r,\,\mathrm{ref}} = \frac{1.400 - 1.300}{2} = 0.050 \text{ in}$$

$$C_D = D_{body,\,ID} - D_{shoulder,\,OD} = 1.400 - 1.400 = 0.000 \text{ in}$$

PLA shoulder test-fit and sanded; extraction ≈19 N (recalled).

## ROUNDED EYELET / LOAD PATH

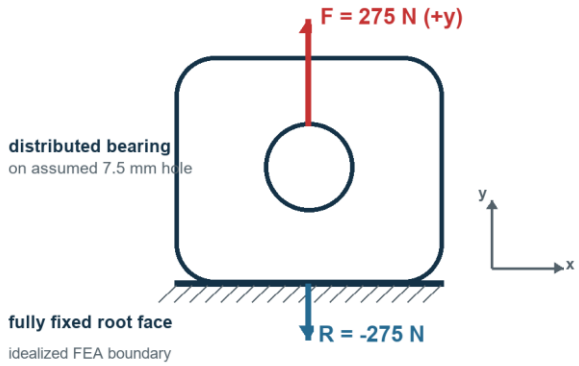


**0.7 × 0.7 × 0.5 in**

Reference envelope

**7.5 mm hole**

Retrospective assumption




## CONDITIONAL RECOVERY SCREEN

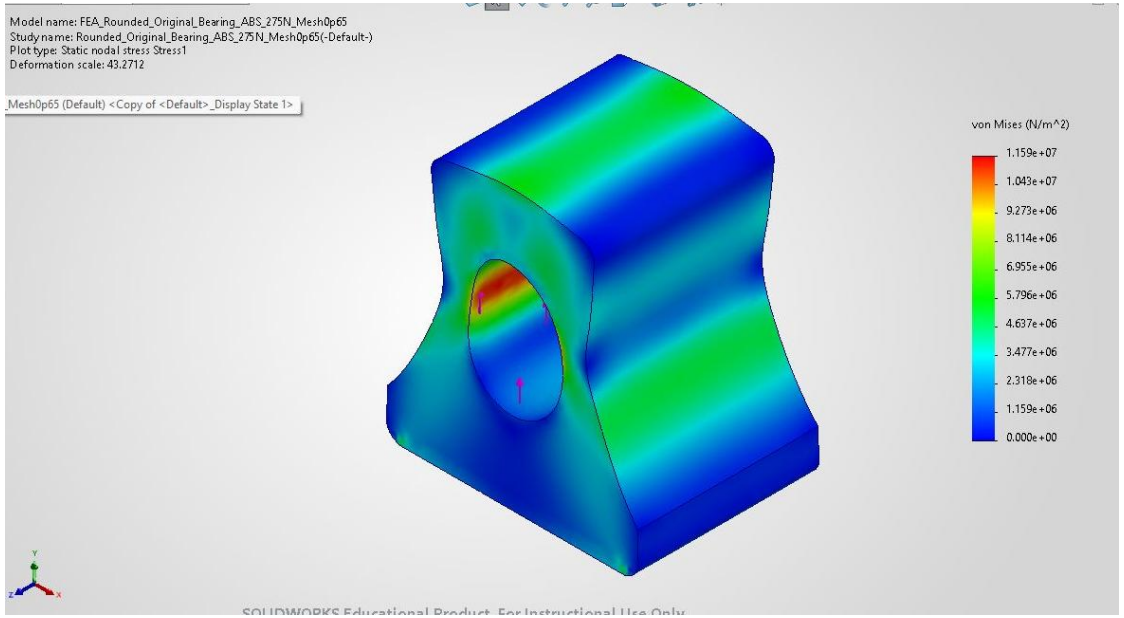
$$F_{\mathrm{open}} = \tfrac{1}{2}\rho V^2 C_D A K_o = 135.5 \text{ N}$$

$$F_{\mathrm{screen}} = K_s F_{\mathrm{open}} = 271 \text{ N} \Rightarrow 275 \text{ N}$$

D = 0.6096 m; ρ = 1.225 kg/m$^3$; Cd = 1.75; t = 1.5 s; opening and screen multipliers = 2 each.

# Wider rounded eyelet lowers modeled stress

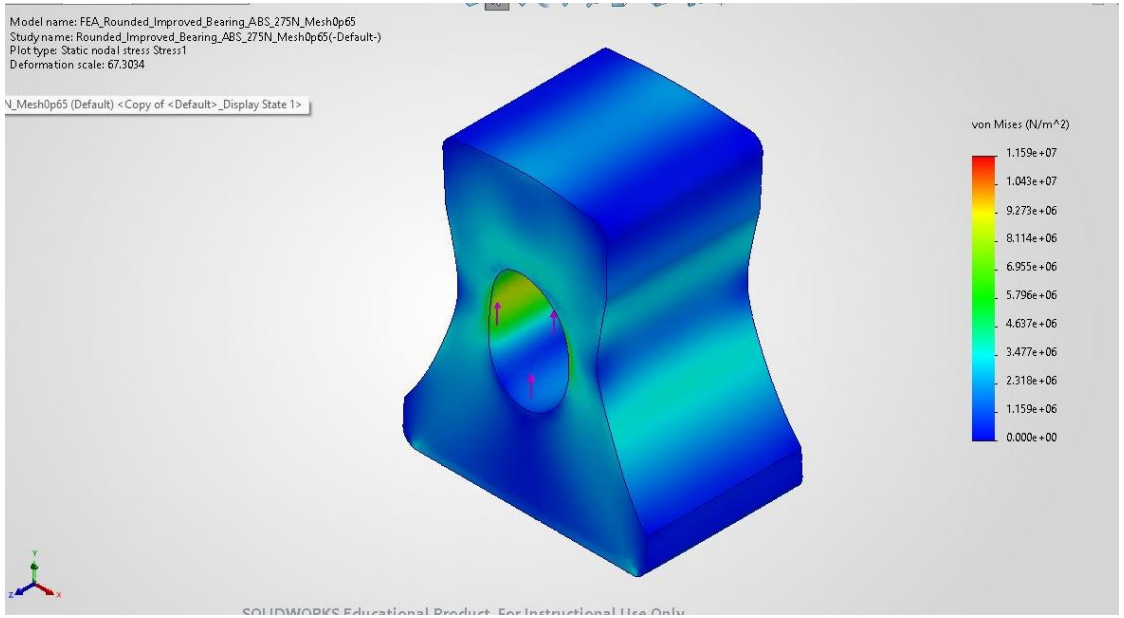
## REFERENCE RECONSTRUCTION




**11.591 MPa → 8.869 MPa**

**−23.5% peak stress**

## WIDER CONCEPT




**0.04138 → 0.03299 mm**

**−20.3% max displacement**

Wider net section and top ligament spread the hole load; added material and an unmodeled root are the tradeoffs.

Common scale 0–11.591 MPa. Conditional 275 N; fixed root; ABS. Printed PLA and dynamics unvalidated.